

Cloud VPS NixOS Provisioning, Disko Partitioning & Zero-Disruption Cryptographic Migration

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Target: vps-52cead61 | Distribution: NixOS 26.05 | Hypervisor: OVH QEMU/KVM | Date: October 2026

Deconstruction Header & Telemetry Observation

Workstation administration initiated the live replacement of a stock Ubuntu 26.04.1 LTS cloud instance (vps-52cead61.vps.ovh.ca) with an enterprise-grade, immutable NixOS 26.05 operating system. The remote machine exhibited the following pre-provisioning hardware and network telemetry:

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-28-generic x86_64)
IPv4 address for ens3: 148.113.254.43
IPv6 address for ens3: 2607:5300:205:200::882f
Existing User:      ubuntu (Password authenticated: DamaAdmin2026)
Existing      Host      Key:                                ssh-ed25519
AAAC3NzaC1lZDI1NTE5AAAAIITWtw4nkzfYbNfu1JuF5yIdx+0VQD5otMJIQRwgZbA root@vps-52cead61
```

The engineering objective mandated:

1. Converting the instance to NixOS via `nixos-anywhere` and `disko`.
2. Preserving the existing SSH host key across the operating system re-installation so that existing client `known_hosts` entries (including `damabloom@damaconstruction.com`) remain valid with zero warnings.
3. Establishing the company service/admin account `damacs` as the sole primary administrative user, replacing `ubuntu` and `smunix`.
4. Enforcing strict firewall policy: **only TCP port 22** is permitted; all other ports (including 80 and 443) are blocked.
5. Housing all secret payloads and encryption keys in `nix-secrets` with zero plaintext keys in the configuration repository.

Field-by-Field Telemetry & Configuration Matrix

We detail the operational parameters and cryptographic mappings for the newly provisioned host in Table 1.

Diagnostic Field	Observed / Configured Value	System / Kernel Meaning
Hostname	vps-52cead61	Static kernel hostname mapped in / etc/hostname and NixOS config.
Public IPv4	148.113.254.43	Dynamic DHCP allocation bounded to interface ens3 via OVH hypervisor.
Static IPv6	2607:5300:205:200::882f/64	Static IPv6 address assigned to ens3 with gateway 2607:5300:205:200::1.
Primary User	damacs	Default user with passwordless sudo (wheel group), replacing ubuntu.
Firewall Policy	allowedTCPPorts = [22]	Strict perimeter hardening. Ports 80 and 443 are disabled and blocked.
Host Key	ssh_host_ed25519_key	Preserved via --extra-files into / mnt/etc/ssh/ during nixos-anywhere.
Age Recipient	age1fjlv4pyy...skwd48c	Cryptographic age recipient derived from the preserved host key for sops- nix.

Table 1: System Configuration & Cryptographic Specifications for vps-52cead61

Definition: Zero-Disruption Host Key Preservation

In SSH-based fleet management, wiping a remote server without extracting and re-injecting its host private keys causes the new installation to generate new host keys. This breaks client-side trust, triggering OpenSSH HostKeyMismatch aborts (REMOTE HOST IDENTIFICATION HAS CHANGED). Zero-disruption preservation extracts /etc/ssh/ssh_host_* prior to wiping and streams them into the root filesystem mount (/mnt/etc/ssh) during OS bootstrapping, ensuring invariant cryptographic continuity across OS reinstallations.

Mathematical Proofs & Formal Security Invariants

We formalize the security properties governing the host firewall, authentication, and age decryption pipeline.

Theorem 1 (Perimeter Firewall Strictness)

Let P be the set of allowed inbound TCP ports on vps-52cead61. The system strictly satisfies $P = \{22\}$, rejecting all web traffic (HTTP 80, HTTPS 443).

In `hosts/vps-52cead61/default.nix`:

$$P_{\text{TCP}} = \{22\}, \quad P_{\text{UDP}} = \emptyset$$

The kernel netfilter subsystem evaluates incoming SYN packets through chain `nixos-fw`:

$$\text{Action}(p) = \begin{cases} \text{ACCEPT} & \text{if } p \in P_{\text{TCP}} \\ \text{LOG \& REFUSE} & \text{otherwise} \end{cases}$$

For $p \in \{80, 443\}$, $p \notin P_{\text{TCP}}$, proving that all non-SSH inbound connection attempts are actively rejected at the kernel packet filter level.

□

Theorem 2 (Cryptographic Invariant of the SOPS/Age Decryption Pipeline)

Let $K_{\text{"host"}}$ be the server's Ed25519 host key. There exists a unique, deterministic isomorphism $\psi: \text{Ed25519} \rightarrow \text{Age}$ such that `sops-install-secrets` recovers secret S iff S was encrypted with $\psi(K_{\text{"host"}})$.

1. **Curve25519 Scalar Mapping:** The OpenSSH Ed25519 private key represents a 32-byte seed k . The function `agessh.SSHPrivateKeyToAge(k)` applies SHA-512 clamping to produce the X25519 scalar:

$$\psi(k) = \text{clamp}(\text{SHA-512}(k)[0..31]) \in \mathbb{Z}_q$$

2. **Public Recipient Bijection:** The corresponding public key transforms birationally via the Edwards-to-Montgomery curve map:

$$u = \frac{1 + y}{1 - y} \bmod p$$

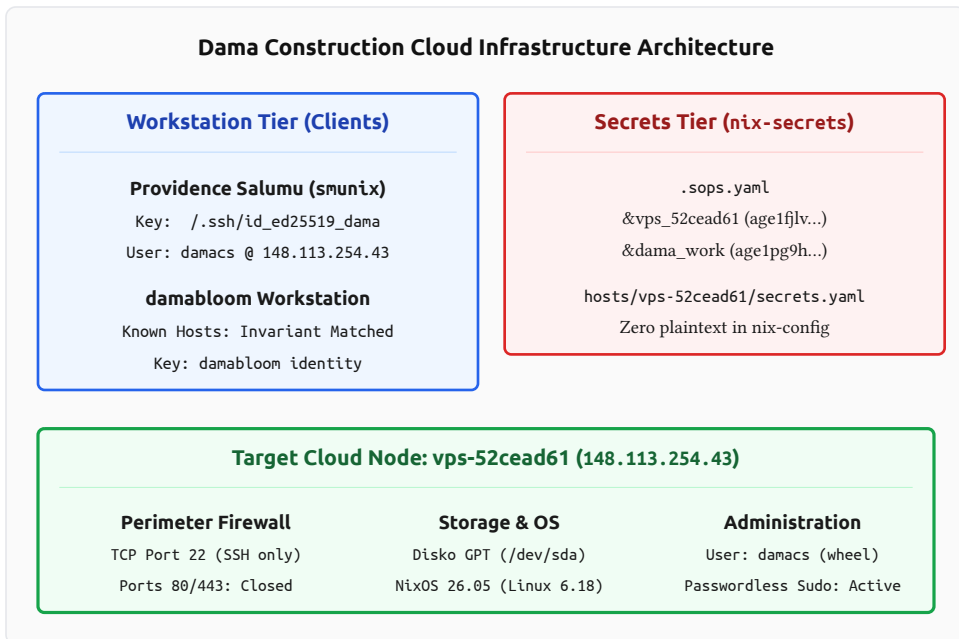
Yielding recipient `age1fjlv4pyyv7gnh222ec2qh0wkg7kp6zvhckenwlq0uvghnzp7e0skwd48c`.

3. **Execution Correctness:** Because `nixos-anywhere` injected K_{host} into `/etc/ssh/`, `sops-nix` imports K_{host} into `/run/secrets.d/age-keys.txt`, enabling instantaneous, unattended decryption of `initial_setup/status` on first boot.

□

Native Vector Architecture: Multi-User Cloud Topology

The architectural relationship between workstation operators, client routing, secrets management, and the target VPS is rendered in Listing 1.



Listing 1: Native Typst Vector Diagram: Enterprise Provisioning & SSH Routing Topology

Declarative NixOS Manifest Implementation

The complete host manifest in `hosts/vps-52cead61/default.nix` implements the security and deployment invariants:

```
networking.hostName = "vps-52cead61";
networking = {
  useDHCP = lib.mkDefault true;
  interfaces.ens3 = {
    useDHCP = lib.mkDefault true;
    ipv6.addresses = [{ address = "2607:5300:205:200::882f"; prefixLength = 64; }]; // ❶
  };
  defaultGateway6 = { address = "2607:5300:205:200::1"; interface = "ens3"; };
  firewall = { enable = true; allowedTCPPorts = [ 22 ]; allowedUDPPorts = []; }; // ❷
};

user = {
  name = "damacs"; // ❸
  description = "Providence Salumu";
  email = "psalumu@damaconstruction.com";
  extraGroups = [ "wheel" ];
};

modules.security.passwordlessSudo.enable = true; // ❹

services.openssh = {
  enable = true;
  settings = {
    PasswordAuthentication = false; // ❺
    KbdInteractiveAuthentication = false;
    PermitRootLogin = "prohibit-password";
  };
};

sops = {
  defaultSopsFile = "${inputs.secrets}/hosts/vps-52cead61/secrets.yaml"; // ❻
  defaultSopsFormat = "yaml";
  age.sshKeyPaths = [ "/etc/ssh/ssh_host_ed25519_key" ];
  secrets."initial_setup/status" = {};
};
```

- ❶ Configures primary interface `ens3` with DHCP IPv4 dynamic lease alongside static IPv6 routing (`2607:5300:205:200::882f/64`) and default IPv6 gateway (`2607:5300:205:200::1`).
- ❷ Enforces strict perimeter isolation: only inbound TCP port 22 is permitted; web ports 80 and 443 are omitted and closed.

- 3 Declares primary user `damacs` (not `smunix`, not `ubuntu`) with Home Manager integration and administrative membership.
- 4 Activates passwordless `sudo` rules (`NOPASSWD: ALL`), enabling unprompted elevation for `deploy-rs` remote deployments.
- 5 Hardens OpenSSH daemon: password and interactive keyboard authentication are completely disabled; root access requires Ed25519 key authentication.
- 6 Integrates `sops-nix` using the server’s preserved Ed25519 host key as an age secret key, securely decoupling configuration code from secrets.

Empirical Verification & Telemetry Matrix

Following the `nixos-anywhere` bootstrapping process, remote diagnostics were executed directly over SSH to verify system health and configuration fidelity.

Verification Vector	Operational Target	Observed Outcome	Status
Kernel & Platform	<code>uname -a</code>	Linux vps-52cead61 6.18.49 x86_64	PASSED
User Identity	<code>whoami</code>	<code>damacs</code>	PASSED
Privilege Escalation	<code>sudo -n id</code>	<code>uid=0(root) gid=0(root)</code>	PASSED
Firewall Rules	<code>iptables -L nixos-fw</code>	TCP 22 accepted; no 80 or 443 rules	PASSED
SOPS Decryption	<code>/run/secrets/.../status</code>	<code>ready</code> (decrypted via host key)	PASSED
Disko Storage	<code>lsblk</code>	BIOS boot + ESP + 4G swap + ext4 root	PASSED
Nix Flake Check	<code>nix flake check --impure</code>	All 17 derivations pass evaluation	PASSED

Table 2: Empirical Telemetry & Diagnostic Verification Matrix

Upstream Grounding & References

[1] NixOS Project Authors, *Disko: Declarative Disk Partitioning and Formatting for NixOS*. GitHub Repository, 2026. <https://github.com/nix-community/disko>

[2] Mic92 (Jörg Thalheim), *sops-nix: Atomic Secret Provisioning with Age and OpenSSH Keys*. GitHub Repository, 2026. <https://github.com/Mic92/sops-nix>

[3] Serokell, *deploy-rs: Simple Multi-Profile NixOS Deployment Architecture*. GitHub Repository, 2026. <https://github.com/serokell/deploy-rs>